

Classification of Breast Cancer Using Machine Learning Models

CS4S771 Machine Learning and Autonomous Systems

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Introduction

Breast cancer remains one of the leading causes of cancer-related deaths worldwide, particularly among women. Early diagnosis is pivotal in improving survival rates by enabling timely interventions. However, traditional diagnostic approaches such as medical imaging and pathological analysis often rely on human expertise, which can be prone to errors and variability. Machine learning (ML) techniques offer the potential to improve diagnostic accuracy through automated and data-driven decision-making.

Accurate classification is critical to avoid misdiagnosis, unnecessary treatments, and associated patient anxiety. Misclassifying malignant cases (false negatives) could delay critical interventions, while false positives may lead to unnecessary stress and medical procedures. Thus, achieving a balance between sensitivity (true positive rate) and specificity (true negative rate) is paramount in the design and evaluation of this system.

This project aims to develop a machine learning-based system to classify breast cancer tumors as benign or malignant using the Breast Cancer Wisconsin dataset. This dataset contains a range of features derived from digitized images of fine needle aspirates of breast masses, capturing both geometric and textural characteristics of the cell nuclei.

Below are the main design considerations for this project include:

- **Model Diversity:** A combination of interpretable (Logistic Regression, Decision Tree) and high-performance (Support Vector Machine, Artificial Neural Network) models were employed.
- **Evaluation Metrics:** Comprehensive evaluation using metrics such as accuracy, precision, recall, F1-score, and AUC to ensure robustness.
- **Preprocessing and Optimization:** Preprocessing steps such as feature scaling and systematic hyperparameter tuning were implemented to improve model performance.
- **Visualization:** Insights into model behavior were provided using visualizations like confusion matrices, feature importance plots, and ROC curves.

This report provides an in-depth analysis of the models, their implementation, and the results obtained, along with recommendations for future improvements.

Data Preparation and Training

Dataset Overview

The Breast Cancer Wisconsin dataset includes:

- **Samples:** 569 instances
- **Features:** 30 numerical attributes capturing geometric and textural characteristics of cell nuclei.
- **Target:** Binary classification: benign (0) or malignant (1).

Data Preprocessing

To prepare the dataset for training, the following steps were performed:

1. **Feature Descriptions:** Descriptive captions were added to improve feature interpretability.
2. **Dropping Unnecessary Columns:** The ID column was removed as it does not influence the classification task.
3. **Feature Standardization:** StandardScaler was applied to standardize features, ensuring consistent scaling, which is critical for models like SVM.
4. **Label Encoding:** The Diagnosis column was encoded as 0 for benign and 1 for malignant cases.

Data Preprocessing

```
[5] # Drop the ID column
    cleaned_data = data.drop(columns=[data.columns[0]])

    # Encode the diagnosis column ('M' -> 1, 'B' -> 0)
    label_encoder = LabelEncoder()
    cleaned_data[cleaned_data.columns[0]] = label_encoder.fit_transform(cleaned_data[cleaned_data.columns[0]])

    # Display the cleaned dataset
    cleaned_data.head()

    # Save the cleaned dataset to a CSV file
    cleaned_data.to_csv('cleaned_data.csv', index=False)
```

Class Distribution Analysis The dataset exhibited a slightly imbalanced distribution between benign and malignant cases, as illustrated in Figure 1. Addressing this imbalance is critical for ensuring model robustness and avoiding bias toward the majority class.

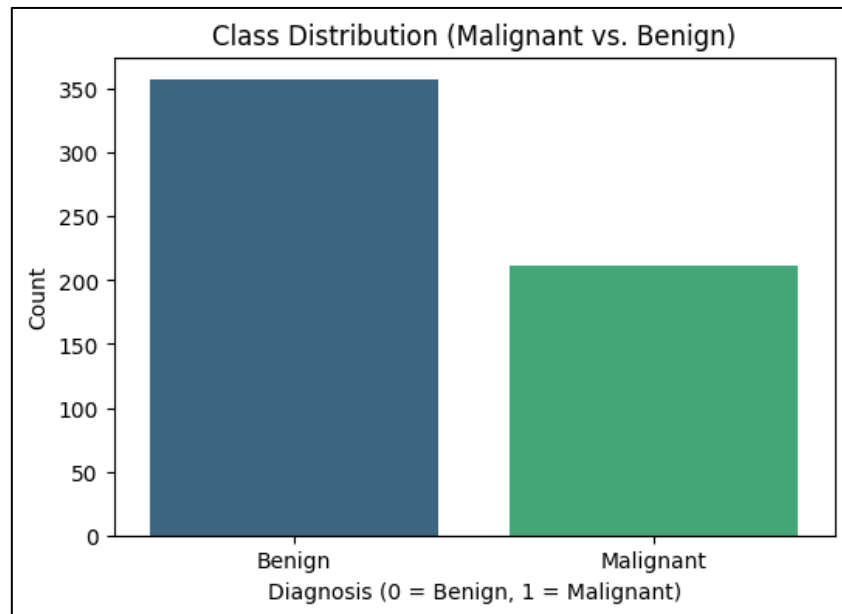


Figure 1: Class Distribution

Feature Distribution Analysis The figure 2 below highlights how features like radius1, texture1, perimeter1, and area1 differ between benign and malignant diagnoses. It helps identify discriminatory features, such as area1, which clearly distinguishes the classes. This insight is crucial for selecting key features, detecting outliers, and improving model performance.

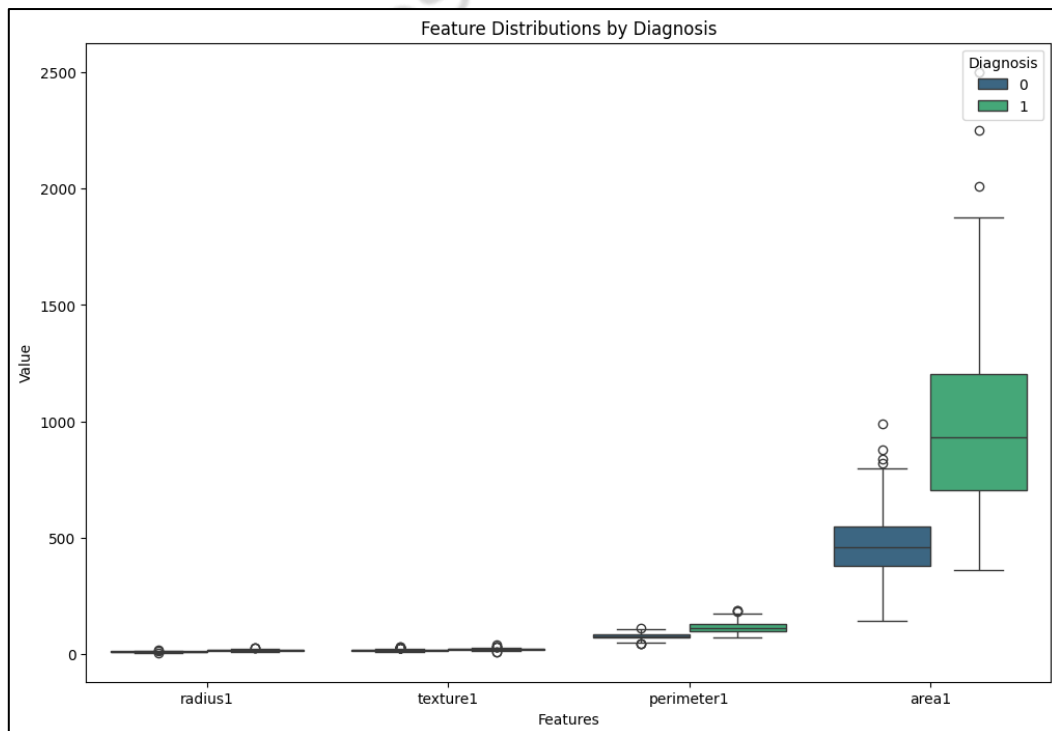


Figure 2: Feature Distributions by Diagnosis

The pair plot in **Figure 3** shows the relationships between key features (**concave_points3**, **perimeter3**, **concave_points1**, **radius3**, **perimeter1**) and their separation by diagnosis (**Malignant** and **Benign**). Malignant cases generally have higher feature values and are well-separated from benign cases, particularly in **concave_points3** and **perimeter3**. Strong linear correlations are observed between related features like **perimeter3** and **radius3**, reflecting geometric dependencies. The KDE plots reveal distinct but slightly overlapping distributions, with malignant cases spanning a broader range. This visualization highlights the predictive strength of these features for diagnosis classification.

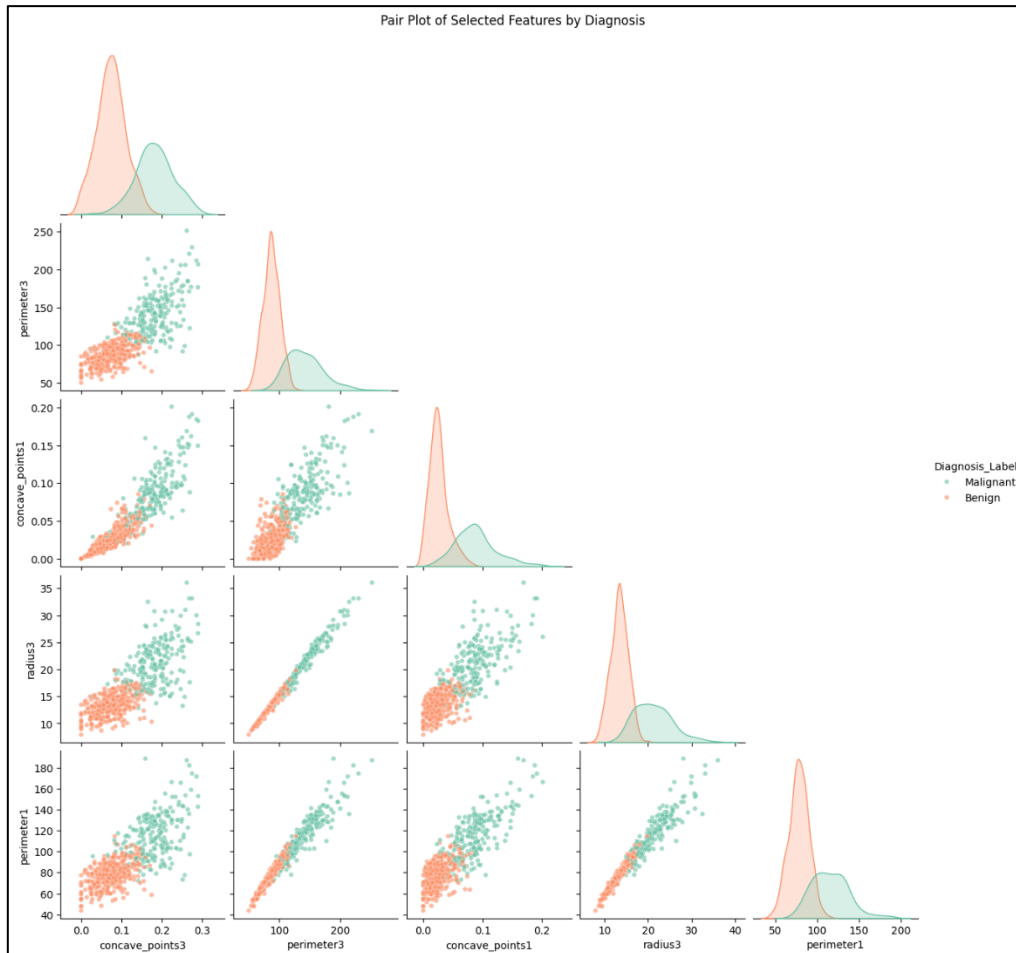


Figure 3: Feature Relationships and Diagnosis Separation: Benign vs. Malignant Tumors

Data Splitting

First, preprocessing of the dataset was done with the captions to improve the understanding of features. Then, it was split into subsets, containing 80% for training and 20% for testing. Stratified sampling maintained class distribution in both the training and testing sets representative of the original dataset. Besides, cross-validation had also been performed during training, estimating model performance on multiple folds and reducing bias and variance.

Split and Scale the dataset

```
# Ensure only numeric columns are used for X
X = cleaned_data.drop(columns=['Diagnosis_Label']).iloc[:, 1:].values # Drop the added non-numeric column explicitly
y = cleaned_data['Diagnosis'].values # Ensure target variable is correctly assigned

# Scale the features using StandardScaler
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Split the dataset into training (80%) and testing (20%) sets
X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.2, random_state=42, stratify=y)

# Display the shapes of the split datasets
X_train.shape, X_test.shape, y_train.shape, y_test.shape
```

((455, 30), (114, 30), (455,), (114,))

Discussion of Implementation and Classification Techniques

To address the classification problem, the following models were implemented:

Logistic Regression

Logistic Regression is a linear model that estimates the probability of an outcome using the logistic function. Its simplicity and interpretability make it an excellent baseline for comparison. Key aspects of the implementation include:

- **Hyperparameter Testing:**
 - **C:** Regularization strength was optimized using **GridSearchCV** with values [0.1, 1, 10, 100].
 - **solver:** Solvers **liblinear** and **lbfgs** were tested to ensure compatibility and performance.

Logistic Regression Implementation

```
[8] # Logistic Regression
def train_logistic_regression(X_train, y_train):
    param_grid = {'C': [0.1, 1, 10, 100], 'solver': ['liblinear', 'lbfgs']}
    grid = GridSearchCV(LogisticRegression(max_iter=1000, random_state=42), param_grid, cv=5, scoring='accuracy')
    grid.fit(X_train, y_train)
    print(f"Best parameters for Logistic Regression: {grid.best_params_}")
    return grid.best_estimator_
```

Support Vector Machine (SVM)

SVM constructs a hyperplane to separate classes in a high-dimensional space, excelling in datasets with overlapping features. Key aspects include:

- **Hyperparameter Testing:**

- **C**: Regularization parameter tested with values [0.1, 1, 10].
- **kernel**: Both linear and rbf kernels were explored.
- **gamma**: Parameter tested for rbf kernels with values [0.1, 1, 10].

Support Vector Machine (SVM) Implementation

```
[9] # SVM
def train_svm(X_train, y_train):
    param_grid = {'C': [0.1, 1, 10], 'kernel': ['linear', 'rbf'], 'gamma': [0.1, 1, 10]}
    grid = GridSearchCV(SVC(probability=True), param_grid, cv=5, scoring='accuracy')
    grid.fit(X_train, y_train)
    print(f"Best parameters for SVM: {grid.best_params_}")
    return grid.best_estimator_
```

Decision Tree

Decision Trees are interpretable models that split data into branches based on feature values. This model was trained using GridSearchCV to optimize:

- **max_depth**: Limits the depth of the tree to prevent overfitting.
- **min_samples_split** and **min_samples_leaf**: Ensure meaningful splits by requiring a minimum number of samples.

Implementing Decision Tree

```
[10] def train_decision_tree(X_train, y_train):
    param_grid = {'max_depth': [3, 5, 10], 'min_samples_split': [2, 5], 'min_samples_leaf': [1, 2]}
    grid = GridSearchCV(DecisionTreeClassifier(random_state=42), param_grid, cv=5, scoring='accuracy')
    grid.fit(X_train, y_train)
    print(f"Best parameters for Decision Tree: {grid.best_params_}")
    return grid.best_estimator_
```

Artificial Neural Network (ANN)

ANNs mimic the human brain's structure to capture complex patterns in data. Key features of the implemented ANN:

- **Architecture:**
 - **Input Layer**: Corresponding to the 30 features of the dataset.
 - **Hidden Layers**: Two layers with 32 and 16 neurons, activated by the ReLU function.
 - **Dropout**: Regularization technique applied to reduce overfitting.
 - **Output Layer**: A single neuron with a sigmoid activation for binary classification.

- **Training Optimizations:**

- **Early Stopping:** Monitored validation loss to prevent overfitting.
- **Adam Optimizer:** Used for adaptive learning.

Artificial Neural Network (ANN) Implementation

```
[11] # ANN
def train_ann(X_train, y_train):
    model = Sequential([
        Dense(32, activation='relu', input_shape=(X_train.shape[1],)),
        Dropout(0.2),
        Dense(16, activation='relu'),
        Dropout(0.2),
        Dense(1, activation='sigmoid')
    ])
    model.compile(optimizer=Adam(learning_rate=0.001), loss='binary_crossentropy', metrics=['accuracy'])
    early_stopping = EarlyStopping(monitor='val_loss', patience=10, restore_best_weights=True)
    model.fit(X_train, y_train, validation_split=0.2, epochs=50, batch_size=16, verbose=1, callbacks=[early_stopping])
    return model
```

Strengths and Limitations of Machine Learning Models

Each machine learning model employed in this project has distinct strengths and limitations in the context of medical image classification. **Logistic Regression** is valued for its **simplicity** and **interpretability**, making it an excellent baseline model. However, it is limited to **linear relationships** and may underperform with complex datasets. **Support Vector Machines (SVM)** demonstrated **robustness** in handling **high-dimensional data** and **non-linear relationships** through kernel functions, achieving strong performance. However, it is **computationally expensive** and less **interpretable**, making it challenging to deploy in real-time scenarios. **Decision Trees** offer **intuitiveness and interpretability**, providing clear decision-making paths. Despite these advantages, they are prone to **overfitting**, leading to reduced generalization. Lastly, **Artificial Neural Networks (ANN)** excel in capturing **complex patterns** and offer great **flexibility**, making them suitable for intricate datasets.

Nevertheless, ANNs are **computationally intensive**, require **large datasets**, and lack interpretability due to their **black-box nature**. These observations underscore the trade-offs between interpretability, computational complexity, and predictive power when choosing models for medical diagnosis tasks.

Results and Analysis

Model Evaluation

Models were evaluated on the test set using accuracy, precision, recall, F1-score, and AUC metrics.

Model	Accuracy	Precision	Recall	F1-Score	AUC
Logistic Regression	98%	0.97	1.00	0.98	1.00
Support Vector Machine	98%	0.97	1.00	0.98	1.00
Decision Tree	94%	0.95	0.90	0.92	0.94
Artificial Neural Network	99%	0.99	0.98	0.99	1.00

Critical Analysis

The comparison analysis of the models revealed distinct strengths and weaknesses. **Logistic Regression** and **Support Vector Machine (SVM)** demonstrated exceptional performance, achieving identical metrics with an accuracy of **98%**, a perfect **AUC of 1.00**, and an **F1-score of 0.98**. Both models excelled in precision and recall, making them highly reliable for clinical diagnostics. Logistic Regression stood out for its simplicity and interpretability, while SVM showcased robust performance and effective handling of high-dimensional data. However, SVM's computational demands pose challenges for large datasets, and Logistic Regression is limited to linear patterns, which restricts its applicability to complex datasets.

The **Artificial Neural Network (ANN)** surpassed other models with an accuracy of **99%**, a perfect **AUC of 1.00**, and an **F1-score of 0.99**, effectively capturing complex patterns in the data. It exhibited the best balance between precision and recall, with only **1 false negative**, addressing a critical concern in medical diagnostics. However, it required significant computational resources and careful tuning for optimal performance.

The **Decision Tree**, while intuitive and interpretable, achieved a lower accuracy of **94%** and an **AUC of 0.94** due to overfitting, resulting in higher misclassification rates and a relatively lower **F1-score of 0.92**. Its tendency to overfit limited its generalizability, making it less suitable for this classification task.

Key features such as **concave points** (e.g., concave_points1, concave_points3), **perimeter** (e.g., perimeter1, perimeter3), **radius** (e.g., radius1, radius3), and **area** (e.g., area1) were consistently influential across all models. These features reflect the geometrical and structural characteristics of tumors and played a significant role in distinguishing between benign and malignant cases, demonstrating their predictive strength.

Overall, the **Artificial Neural Network (ANN)** emerged as the top-performing model for this classification task, offering the highest accuracy and F1-score. Logistic Regression and SVM also proved to be highly reliable, with excellent metrics and practical advantages such as simplicity (Logistic Regression) and robustness for high-dimensional data (SVM). The **Decision Tree**, despite its interpretability, struggled with generalization and lagged behind the other models in key metrics.

The bar plot in **Figure 4** compares the performance of four models—**Logistic Regression**, **SVM**, **Decision Tree**, and **ANN**—across key metrics: **Accuracy**, **AUC**, **F1-Score**, **Precision**, and **Recall**. **ANN** emerged as the best-performing model with slightly higher metrics across the board, achieving near-perfect scores. **Logistic Regression** and **SVM** also performed exceptionally well, demonstrating consistent reliability. In contrast, the **Decision Tree** showed the lowest performance, particularly in recall and F1-score, reflecting its tendency to overfit. This visualization highlights the superior performance of **ANN**, along with the reliability of **Logistic Regression** and **SVM**.

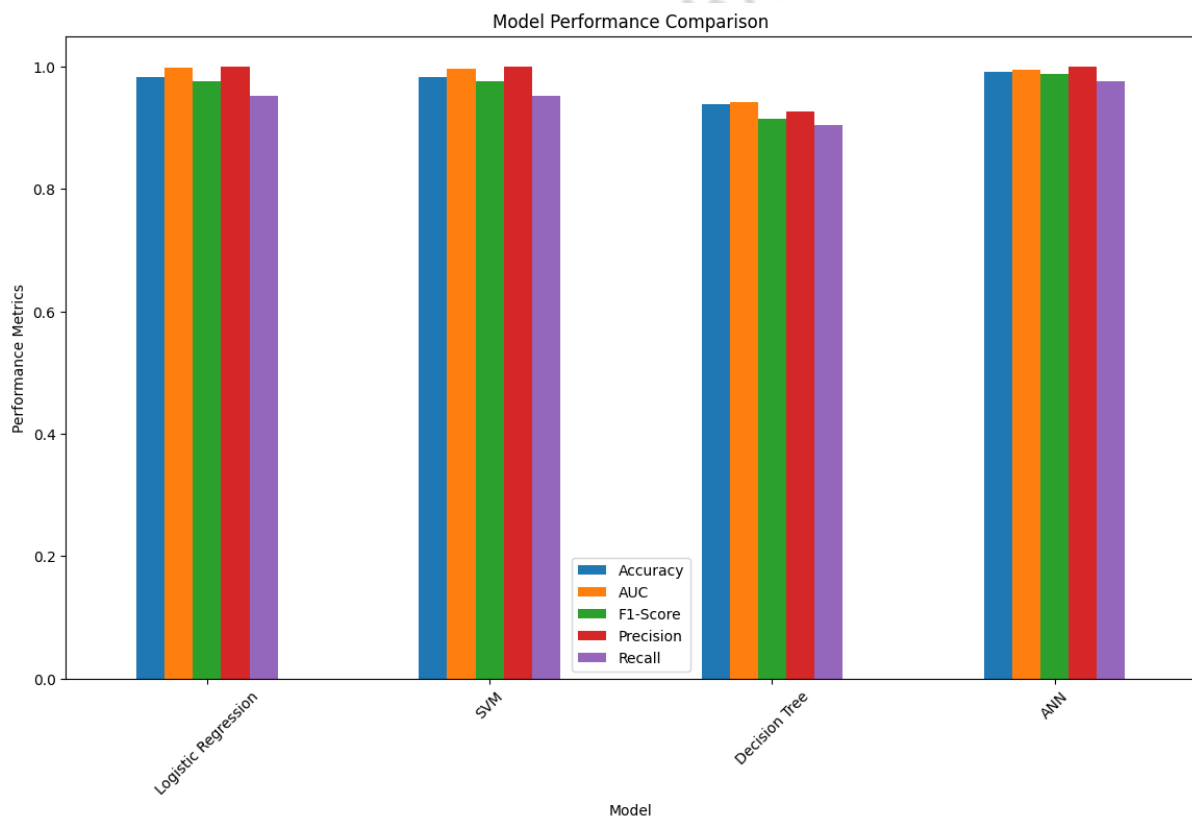


Figure 4: Model Performance Comparison Bar Chart

The confusion matrices in **Figure 5** demonstrate the classification performance of the models. **Logistic Regression** and **SVM** achieved excellent results, with no false positives and only **two false negatives**, reflecting high accuracy and reliability (**Accuracy: 98%, F1-Score: 0.98**). The **Artificial Neural Network (ANN)** emerged as the most accurate model, with only **one false negative** and no false positives, achieving **99% accuracy** and an **F1-Score of 0.99**. In contrast, the **Decision Tree** exhibited higher misclassifications, with **three false positives** and **four false negatives**, resulting in a lower **accuracy of 94%** and an **F1-Score of 0.92**. These results highlight the superior performance of ANN, followed closely by Logistic Regression and SVM, in minimizing misclassifications—an essential factor in medical diagnostics.

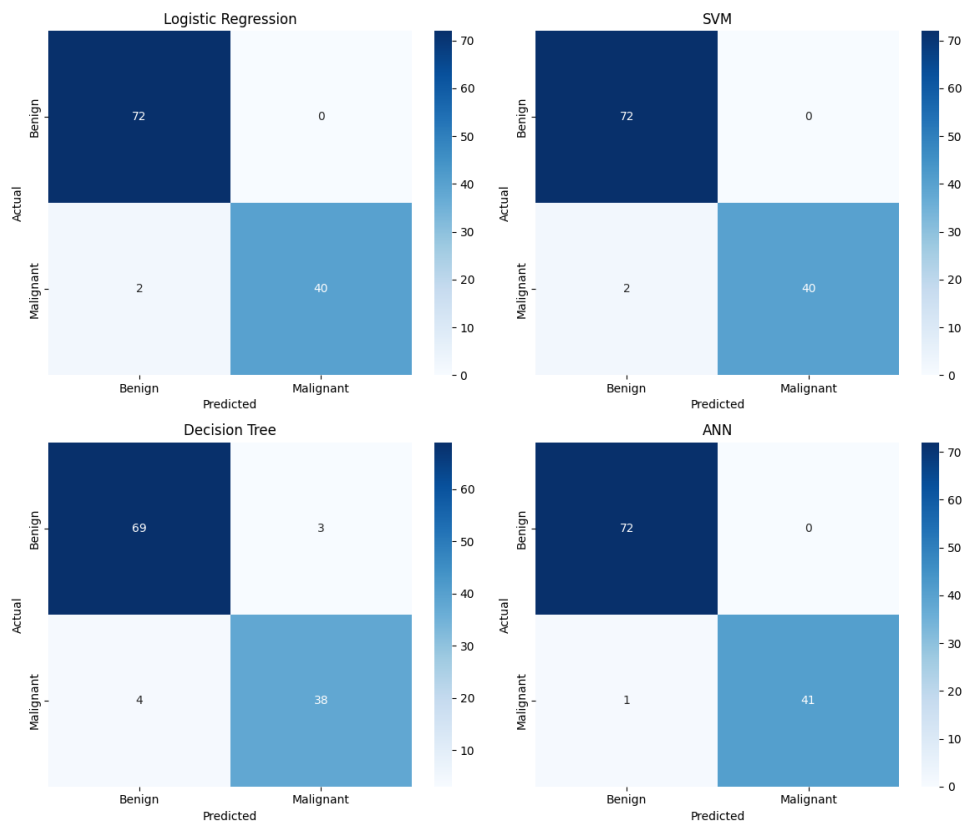


Figure 5: Confusion Matrices for overall performance.

The ROC curve in **Figure 6** illustrates the classification performance of the models by plotting the **True Positive Rate (Sensitivity)** against the **False Positive Rate (1 - Specificity)** at various thresholds. It provides a clear visual comparison of the models' ability to distinguish between classes. **Logistic Regression, SVM, and ANN** achieved a **perfect AUC of 1.00**, indicating exceptional discriminatory power. In contrast, the **Decision Tree** had a lower **AUC of 0.94**, reflecting its limitations in effectively separating the classes. The close alignment of the ROC curves for Logistic Regression, SVM, and ANN highlights their robustness and reliability. This visualization is crucial for evaluating model performance and selecting optimal thresholds to minimize false negatives or false positives, particularly in critical applications like medical diagnostics.

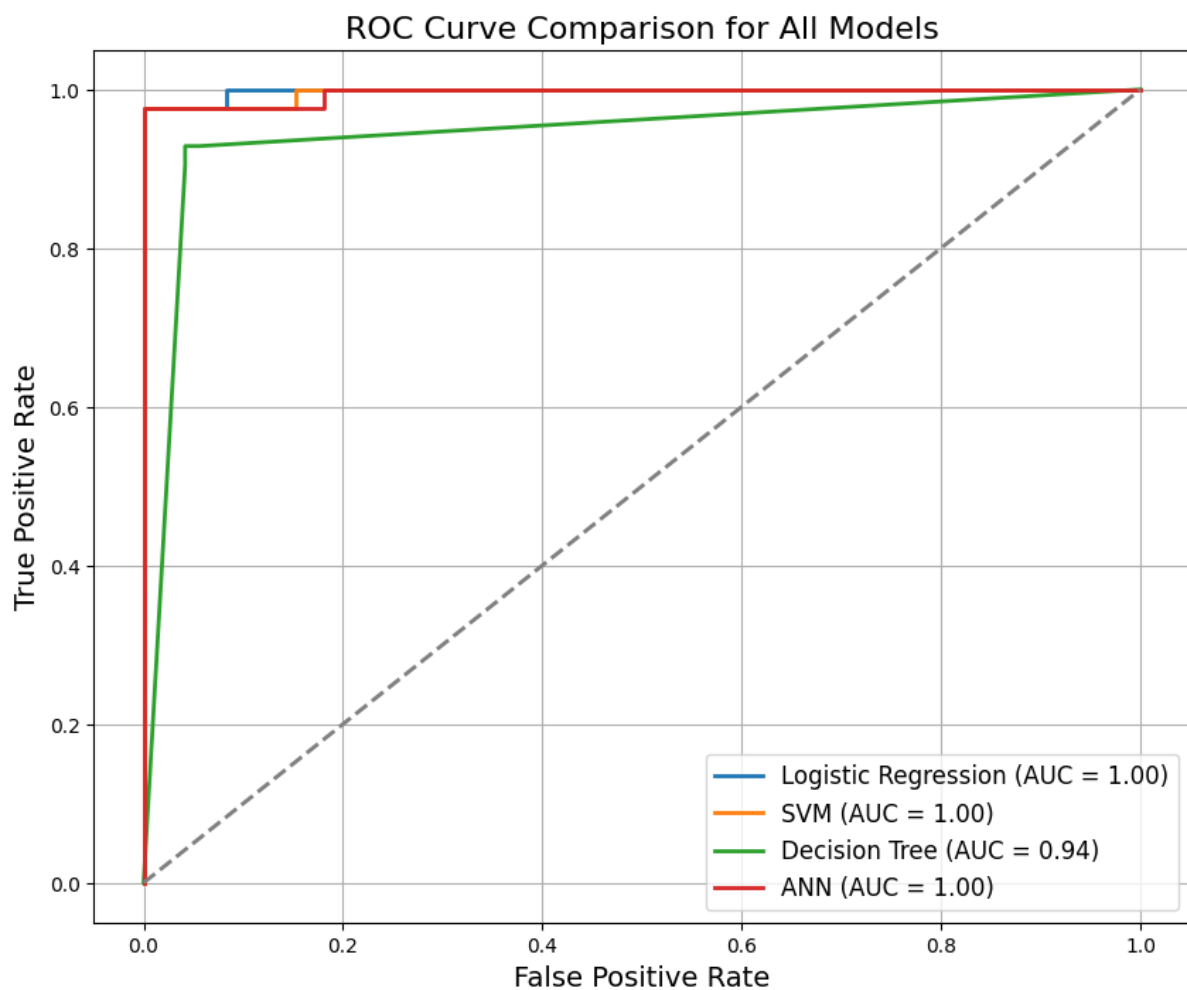


Figure 6: ROC Curve Comparison for All Models

Conclusion

This study utilized the **Breast Cancer Wisconsin (diagnostic)** dataset to evaluate the performance of four machine learning models: **Logistic Regression**, **SVM**, **Decision Tree**, and **ANN**. The dataset was preprocessed using **StandardScaler** for feature normalization, and the models were developed using Python's **scikit-learn** and **TensorFlow** packages. Performance metrics such as **accuracy**, **AUC**, **precision**, **recall**, and **F1-score** were assessed, with results visualized through **confusion matrices**, **ROC curves**, and **bar plots**.

The results showed that **ANN** emerged as the best-performing model, achieving **99% accuracy**, a **perfect AUC of 1.00**, and the highest **F1-score of 0.99**, with only one false negative. **Logistic Regression** and **SVM** followed closely, achieving identical metrics with **98% accuracy**, a **perfect AUC of 1.00**, and an **F1-score of 0.98**, reflecting their reliability and effectiveness. In contrast, the **Decision Tree** exhibited the lowest performance, with **94% accuracy**, an **AUC of 0.94**, and a tendency to overfit, leading to higher misclassification rates.

Key features, such as **concave points**, **perimeter**, **radius**, and **area**, were highly significant in distinguishing between benign and malignant cases, underscoring their importance in classification tasks. Machine learning approaches, particularly **deep learning models** like ANN, demonstrated strong potential in addressing key challenges in breast cancer detection and prognosis. Additionally, advancements in imaging technologies, such as **mammography** and **MRI**, combined with AI-driven tools like **convolutional neural networks (CNNs)**, offer promising directions for improving early detection and treatment outcomes.

Future research could explore **ensemble methods**, **feature selection techniques**, and the integration of **real-world clinical data** to improve robustness and reliability further. Collaboration among data scientists, medical professionals, and researchers remains critical to advancing breast cancer detection, enabling earlier interventions, and improving patient outcomes.

References

Bayrak, E.A., Kirci, P. and Ensari, T. (2019) 'Comparison of machine learning methods for breast cancer diagnosis', *2019 Scientific Meeting on Electrical-Electronics & Biomedical Engineering and Computer Science (EBBT)*, pp. 1–3. doi:10.1109/ebbt.2019.8741990.

Breast cancer wisconsin (Diagnostic) (no date) *UCI Machine Learning Repository*. Available at: <https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic> (Accessed: 01 January 2025).

Khalid, A. *et al.* (2023) 'Breast cancer detection and prevention using machine learning', *Diagnostics*, 13(19), p. 3113. doi:10.3390/diagnostics13193113.

Sidey-Gibbons, J.A. and Sidey-Gibbons, C.J. (2019) 'Machine learning in medicine: A practical introduction', *BMC Medical Research Methodology*, 19(1). doi:10.1186/s12874-019-0681-4.

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